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1 October 2026

Describe computer cabling

US 14929

Section A: FORMATIVE ASSESSMENT

QUESTION 1

i. Different types of computer cabling

1. Twisted Pair Copper Cable - Most common for LAN. Two types:

   - UTP (Unshielded Twisted Pair) - Cat5e, Cat6, Cat6a

   - STP (Shielded Twisted Pair) - with foil shield

   - Two forms: Solid copper (for backbone) and Stranded copper (for patch leads)

2. Coaxial Copper Cable

   - RG6, RG59, RG11 - 75 ohm for TV/CCTV

   - RG58 - 50 ohm for old Ethernet (10Base2)

3. Fibre Optic Cable

   - Single-mode (OS1, OS2) - long distance

   - Multi-mode (OM1, OM2, OM3, OM4, OM5) - short distance LAN

ii. Briefly explain history of each cabling system

- Coaxial: invented 1880 by Oliver Heaviside. Used as first Ethernet cable in 1970s-80s - Thicknet 10Base5 and Thinnet 10Base2, speed 10Mbps. Replaced by twisted pair in 1990s but still used today for DSTV and CCTV.

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- Twisted Pair: invented 1881 by Alexander Graham Bell for telephone to reduce interference (crosstalk). Adopted for networks in 1990 with 10Base-T standard. Then Cat3 (10Mbps), Cat5 (100Mbps 1995), Cat5e (1Gbps 2001), Cat6 (1Gbps-10Gbps 2002), Cat6a/Cat7/Cat8 now for 10-40Gbps.

- Fibre Optic: Developed 1970 by Corning Glass - uses light through glass instead of electricity. First field test 1977. Used for telecom backbone in 1980s, now for LAN backbone, FTTH, data centres because it is very fast, long distance, no EMI.

QUESTION 2

Describe current and future trends in cabling systems

i. Characteristics

- Current: High speed 1Gbps to 10Gbps, low interference, need for Power over Ethernet (PoE), supports both synchronous (video streaming) and asynchronous (email) data, more secure than wireless.

- *Future:* Ultra-high speed 25Gbps/40Gbps/100Gbps with Cat8 and OM5 fibre, smart cabling that powers devices (PoE++ 90W), integrated with IoT and smart buildings, less downtime, self-diagnosing cables.

ii. Materials

- current: copper (99.99% pure) for twisted pair/coax, glass/plastic core for fibre, PVC or LSZH (Low Smoke Zero Halogen) jacket, aluminium foil and braided shield for STP.

- future: more fibre glass than copper (cheaper long term, lighter), composite fibre, biodegradable jackets, smaller diameter (28AWG) for better airflow in cabinets, copper-clad aluminium being replaced by pure copper due to standards.

iii. Standards thereof

- Current: TIA/EIA-568-C, ISO/IEC 11801, IEEE 802.3 for Ethernet. Categories: Cat5e, Cat6, Cat6a, Cat7. Fibre standards: OS1/OS2 single-mode, OM3/OM4 multi-mode. PoE IEEE 802.3af/at.

- future: Cat8 / Class I & II standard (TIA-568.2-D) for 25/40Gbps, OM5 wideband multi-mode, IEEE 802.3bt PoE++ Type 3 & 4 (60W/90W), green building standards (BICSI), standards for 200G/400G fibre.

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QUESTION 3

Describe termination methods

i. Coaxial copper

- Methods:

1. BNC connector - bayonet twist for old LAN/CCTV, crimped with crimper tool
2. F-type connector - screw type for DSTV, crimp or compression
3. N-type - threaded for outdoor

- characteristics: Must maintain 50 or 75 ohm impedance, needs good contact of braid shield to avoid signal loss, strip carefully (outer jacket, braid, dielectric, centre core).

- Features: Easy and cheap, but if badly terminated causes reflection and loss. Good for both synchronous (continuous video) and asynchronous. Compression F-type is most reliable today.

ii. Fibre optic

- Methods:

1. Epoxy and polish - glue fibre into ST/SC/LC connector and polish end face with puck
2. Mechanical splice / pre-polished splice-on connector (SOC) - cleave fibre and align with gel in clip, quick, no power
3. Fusion splicing - machine melts two fibres together with electric arc - permanent, lowest loss 0.01-0.1dB

- characteristics: Must be extremely clean (use alcohol wipes), no dust, must not bend beyond 30mm radius, uses light not electricity so no EMI, needs special tools (stripper, cleaver, splicer, OTDR tester).

- Features: very high speed, supports long distance, excellent for synchronous high-bandwidth data (video conferencing) and asynchronous. LC connector small form factor is current trend for high density. Fusion splicing is best for backbone.